

ShopSphere

E-Commerce Management System

Software Requirements Specification (SRS)

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Course: Software Engineering (5th Semester)

Date: September 23, 2026

Version: 1.0

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1. Introduction

1.1 Purpose

ShopSphere is an E-Commerce Management Software which is designed to provide customers with an efficient platform to browse, search, and purchase products online. The software also enables administrators to manage products, inventory, orders, payments, and customer information through a centralized system.

1.2 Scope

ShopSphere facilitates online shopping by offering product management, shopping cart functionality, secure payment processing, order tracking, and customer account management. It aims to provide a user-friendly, secure, and scalable e-commerce solution.

1.3 Definitions, Acronyms, Abbreviations:

- SRS: Software Requirement Specification
- DFD: Data Flow Diagram
- COD: Cash on Delivery
- DBMS: Database Management System
- HTTPS: Secure Hypertext Transfer Protocol
- FR : Functional Requirement

2. Overall Description

2.1 Product Perspective

ShopSphere is a standalone web-based system connected to a database and payment gateway. It provides a platform for customers to shop online and for administrators to manage business operations. The system allows customers to register, browse products, add items to a cart, place orders, make online payments, and track their orders. Administrators can manage

products, categories, inventory, customers, and orders. The system ensures secure data management and provides an efficient way to manage online sales.

2.2 Product Functions

The software provides:

- User Registration and Login
- Product Search and Filtering
- Shopping Cart Management
- Online Payment Processing
- Order Management
- Inventory Management
- Customer Support
- Report Generation

2.3 User Classes

Customer :

- Creates account
- Searches products
- Places orders
- Makes payment

Administrator :

- Adds and updates products
- Manages orders
- Monitors inventory

- Generates reports

Guest User :

- Can browse products
- Cannot purchase without registration

2.4 Operating Environment

- Operating System: Windows/Linux
- Browser: Chrome, Firefox, Edge
- Database: MySQL
- Web Server: Apache/Nginx

3. Functional Requirements

FR1: User Registration and Login

The system shall allow users to create an account using email and password. Registered users can log in securely and reset passwords if forgotten.

Input:

- Name
- Email
- Password

Output:

- Successful account creation
- Login confirmation

FR2: Product Management

Administrators can add, edit, remove, and update product details including price, category, images, and stock availability.

Input:

- Product name
- Price
- Category
- Quantity

Output:

- Updated product catalog

FR3: Product Search

The system shall allow customers to search products using keywords, categories, brands, and price ranges.

Input:

- Search keyword

Output:

- Matching product list

FR4: Shopping Cart Management

Customers can add products to the cart, remove items, and update quantities before making a purchase.

Input:

- Product selection

Output:

- Updated cart information

FR5: Order Processing

After checkout, the software will generate orders and maintain order records.

Input:

- Customer details
- Cart details

Output:

- Order confirmation

FR6: Payment Processing

The system shall support online payment methods such as credit cards, debit cards, UPI, and net banking.

Input:

- Payment details

Output:

- Payment success/failure notification

FR7: Inventory Management

The quantity of products shall automatically update whenever a purchase is made.

Input:

- Order information

Output:

- Updated stock information

FR8: Order Tracking

Customers can track order status such as:

- Pending
- Processing
- Shipped
- Delivered

Output:

- Order status details

4. Non-Functional Requirements

4.1 Performance

The software should process user requests quickly and support multiple users simultaneously without significant delays.

4.2 Security

The system shall:

- Encrypt user passwords
- Use HTTPS
- Prevent unauthorized access
- Protect payment information

4.3 Reliability

The software should operate continuously with minimal downtime and recover data during system failures.

4.4 Scalability

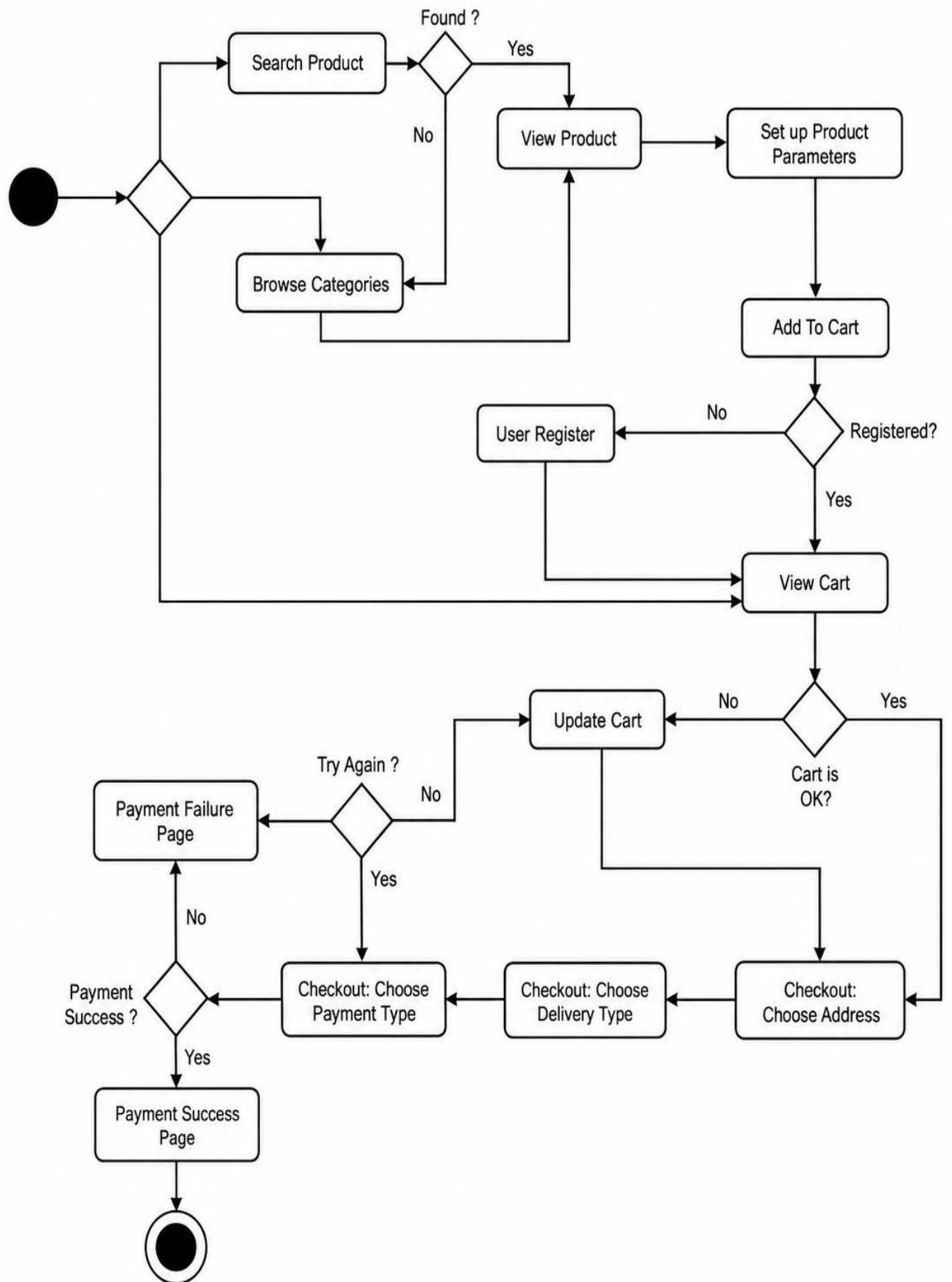
The system should support increasing numbers of users, products, and transactions without major redesign.

4.5 Usability

The interface should be simple, attractive, and easy to use for both technical and non-technical users.

5. Activity Diagram

1. The process starts when a customer enters ShopSphere.
2. The customer searches for a product in the system.
3. If the product is not found, the customer can browse product categories.
4. If the product is found, the customer views the product details.
5. The customer selects the required product parameters and adds the product to the shopping cart.
6. The system checks whether the customer is registered.
7. If the customer is not registered, registration is completed before proceeding.
8. The customer reviews the shopping cart and updates it if required.
9. If the cart is correct, the customer proceeds to checkout.
10. During checkout, the customer selects the delivery address and delivery method.
11. The customer chooses a preferred payment method.
12. The system processes the payment transaction.
13. If the payment fails, the customer can retry the payment or modify the cart.
14. If the payment is successful, the order is confirmed and a success page is displayed.
15. The process ends after successful order placement.



6. Use Case Diagram

6.1 Actors Description :

Customer: Registers/logs into the application, searches and views products, manages the shopping cart, places orders, makes online payments, and tracks order statuses.

Guest User: Browses products and categories; requires account registration before checking out or completing a purchase.

Administrator: Manages product catalog and categories, updates inventory, monitors orders, and generates analytical reports.

Payment Gateway: External actor responsible for authorizing, verifying, and processing payment transactions.

6.2 Use Case Specifications

UC1: Register & Authenticate: Facilitates user sign-up, login, and session validation (extends password recovery).

UC2: Search & Browse Products: Allows querying the catalog by keywords, brands, and categories.

UC3: Manage Cart: Includes adding items, modifying quantities, and removing products from the active session.

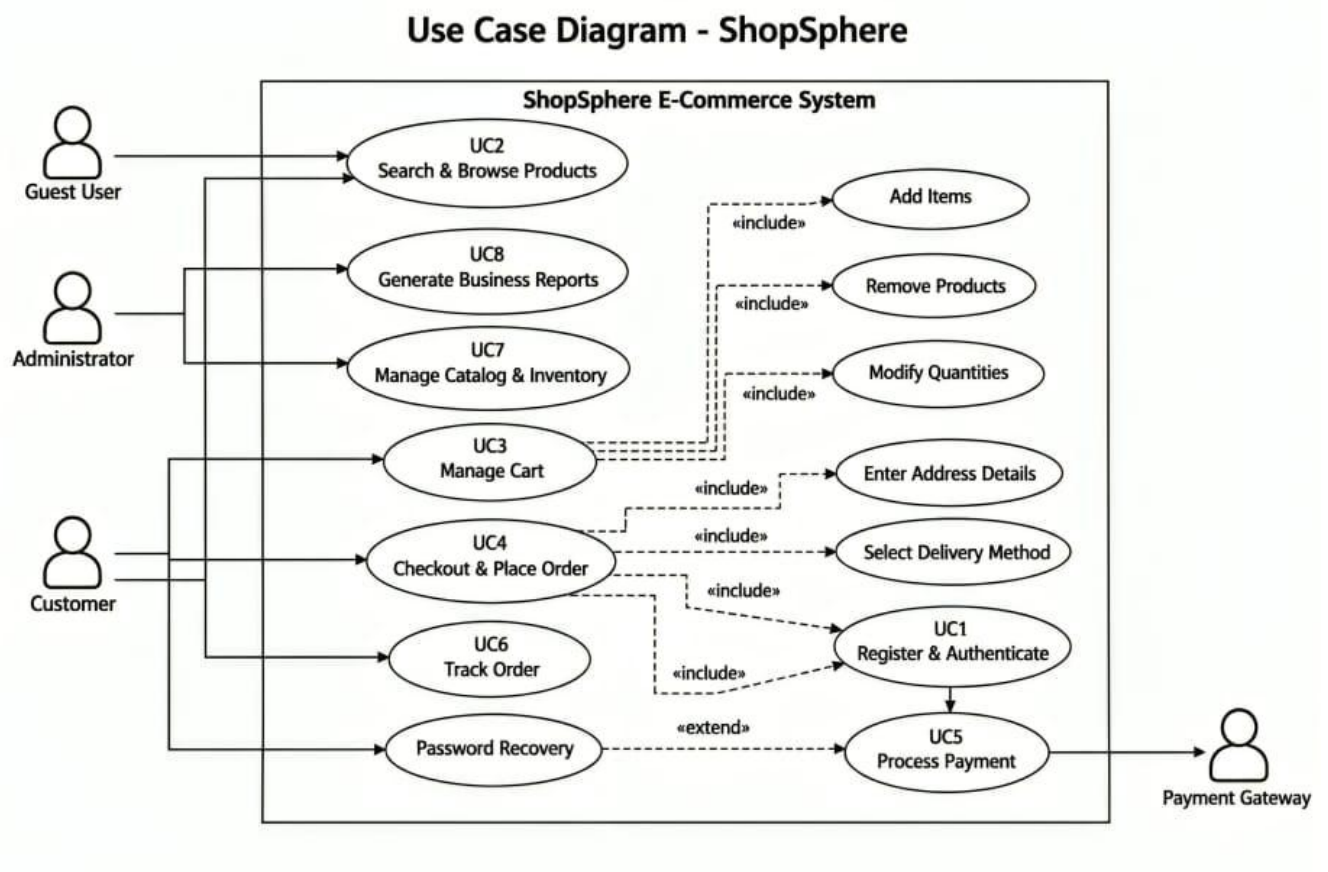
UC4: Checkout & Place Order: Collects address details, delivery method, and triggers payment processing.

UC5: Process Payment: Coordinates with the Payment Gateway to verify transactions and confirm order placement.

UC6: Track Order: Provides order status tracking (Pending, Processing, Shipped, Delivered).

UC7: Manage Catalog & Inventory: Allows administrators to add, edit, or remove items and track inventory levels.

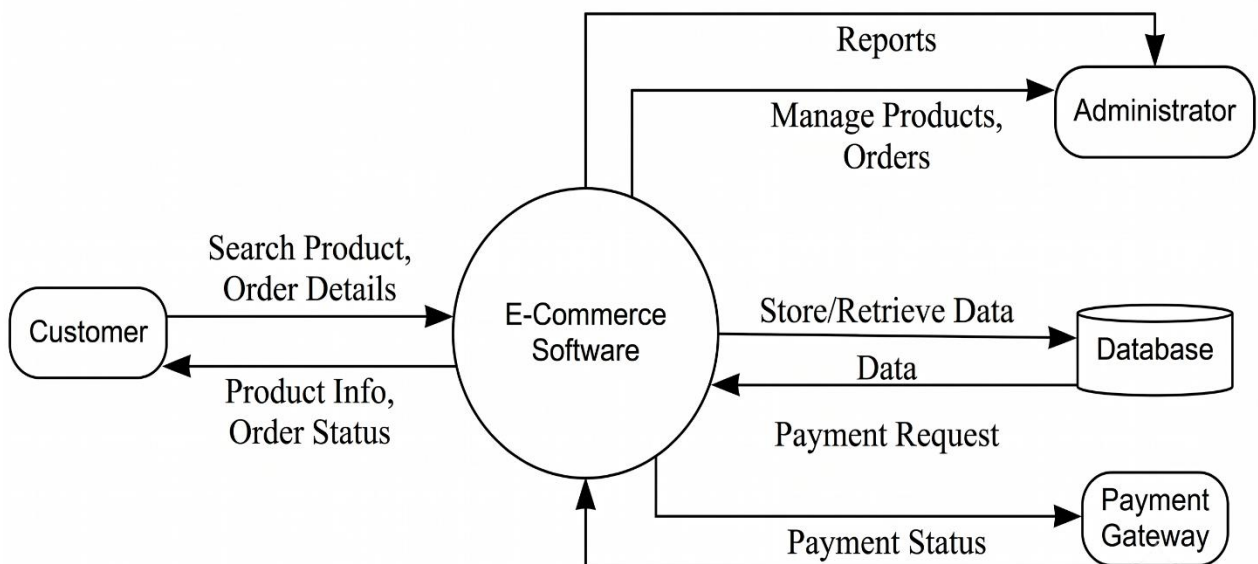
UC8: Generate Business Reports: Produces system sales and order reports for administrative review.



7. Data Flow Diagrams

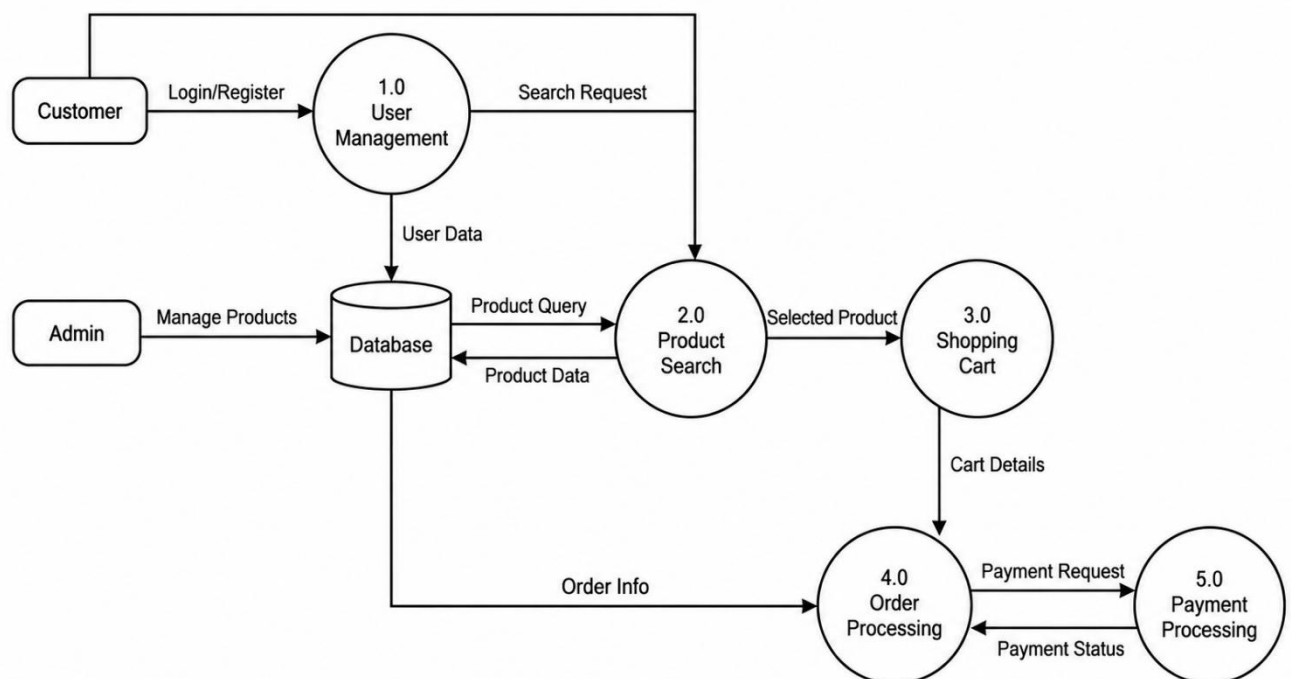
7.1 DFD Level 0 Description :

1. DFD Level 0 provides a high-level view of the E-Commerce Management Software(ShopSphere).
2. It shows the interaction between the Customer, Administrator, Payment Gateway, and the system.
3. Customers can search products, place orders, make payments, and track orders through the system.
4. Administrators manage products, inventory, and reports.
5. The Payment Gateway processes online transactions and returns payment status information.



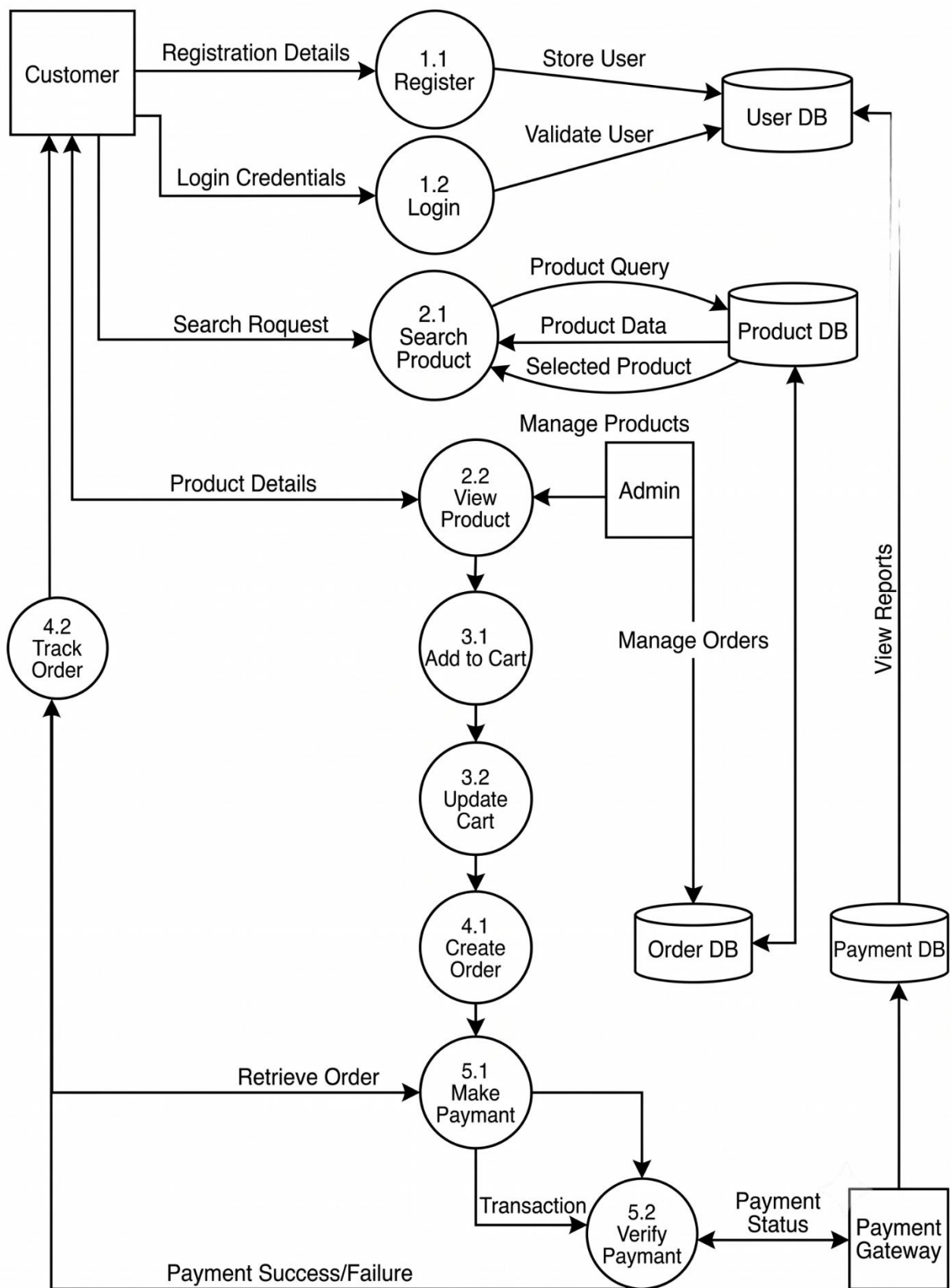
7.2 DFD Level 1 Description :

1. DFD Level 1 breaks down the E-Commerce Management Software(ShopSphere) into its major functional modules.
2. The User Management module handles customer registration and login activities.
3. The Product Search module allows customers to search and view available products.
4. The Shopping Cart module manages selected products before checkout.
5. The Order Processing module creates and maintains customer orders.
6. The Payment Processing module handles online payment transactions and updates payment status.
7. The Administrator manages products and inventory through the system database.
8. The modules work together to complete the online shopping process efficiently.



7.3 DFD Level 2 Description :

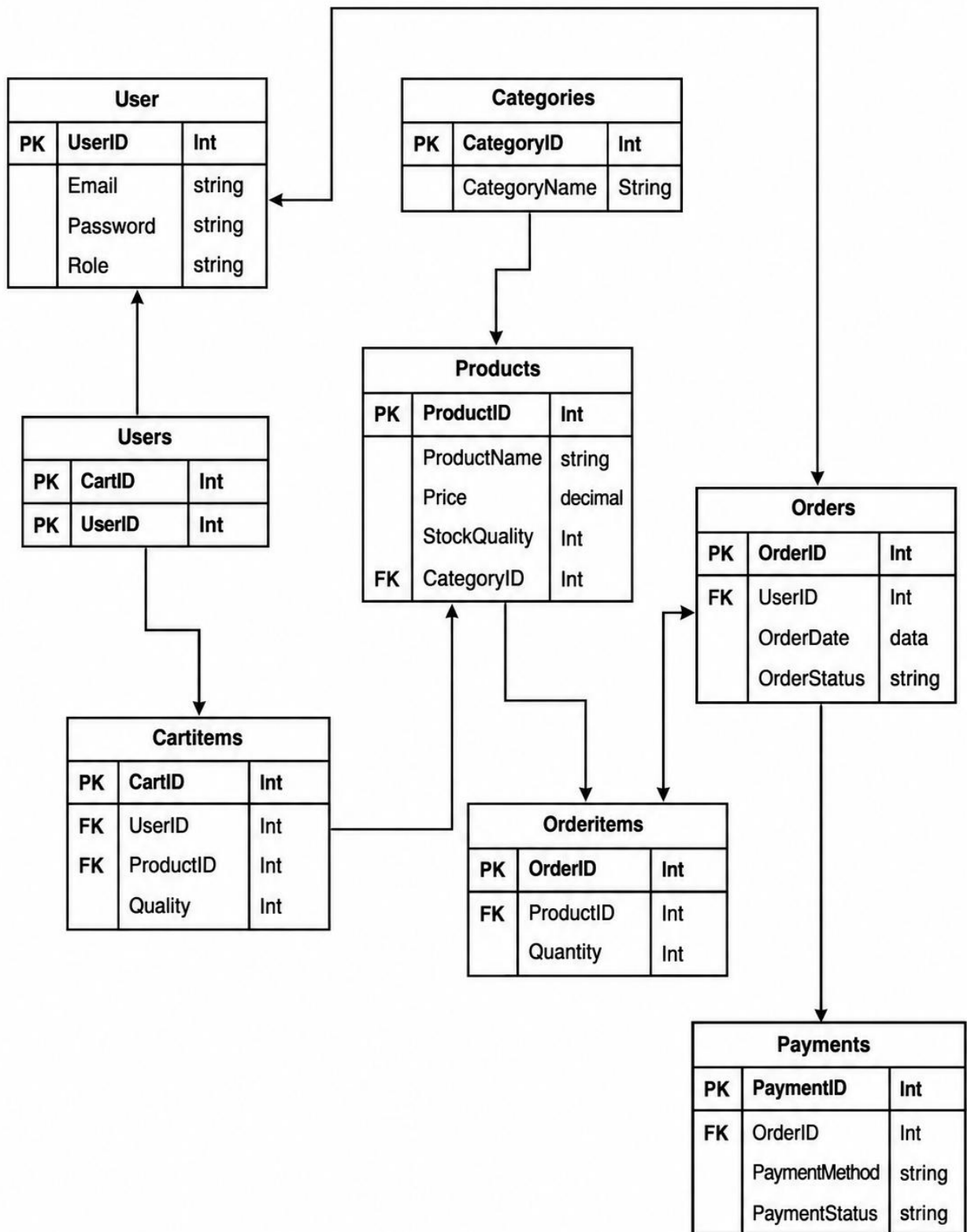
1. DFD Level 2 provides a detailed representation of the internal processes of the E-Commerce Management Software(ShopSphere).
2. The User Management process is divided into Registration and Login Verification sub-processes, which validate user information and store records in the user database.
3. The Product Search process is divided into Product Search and Product View operations, enabling customers to browse products and view detailed information.
4. The Shopping Cart process includes Add to Cart and Update Cart functions, allowing customers to manage selected products before purchase.
5. The Order Processing process consists of Order Creation and Order Tracking activities, which generate order records and provide order status information to customers.
6. The Payment Processing module is further divided into Payment Transaction and Payment Verification processes, which communicate with the Payment Gateway to authorize payments.
7. Different databases are used to store user details, product information, order records, and payment details.
8. Administrators can manage products, monitor orders, update inventory, and generate business reports using the stored data.
9. This level explains the detailed flow of information between processes, databases, customers, administrators, and the payment gateway.



8. Entity Relationship Diagram

8.1 Core Entities & Attributes :

- User: Stores authentication and access levels.
 - Attributes: UserID (PK), Email, Password, Role.
- Categories: Groups inventory items.
 - Attributes: CategoryID (PK), CategoryName.
- Products: Manages item catalog details.
 - Attributes: ProductID (PK), ProductName, Price, StockQuantity (Quantity), CategoryID (FK).
- Cart (Users): Represents an active user shopping cart session.
 - Attributes: CartID (PK), UserID (PK/FK).
- CartItems: Stores individual items within a shopping cart.
 - Attributes: CartID (PK), UserID (FK), ProductID (FK), Quantity (Quantity).
- Orders: Records completed user purchases.
 - Attributes: OrderID (PK), UserID (FK), OrderDate, OrderStatus.
- OrderItems: Line-item breakdown linking orders to specific products.
 - Attributes: OrderID (PK), ProductID (FK), Quantity.
- Payments: Handles billing and payment gateway records.
 - Attributes: PaymentID (PK), OrderID (FK), PaymentMethod, PaymentStatus.



8.2 Relationships & Cardinality :

1. User to Cart & Orders: A User can maintain a shopping cart and place multiple Orders (1-to-Many), mapped via UserID.
2. Category to Products: Each Category classifies multiple Products (1-to-Many), linked by CategoryID.
3. Cart & Products to CartItems: CartItems acts as an associative entity linking a user's cart session and Products (Many-to-Many resolution).
4. Orders & Products to OrderItems: OrderItems resolves the Many-to-Many link between Orders and Products, tracking purchased quantities.
5. Orders to Payments: Each Order links directly to its transaction via OrderID in Payments.

9. Database Design

9.1 User Table

Field Name	Data Type	Description
UserID	INT	Primary Key
Email	VARCHAR(100)	User Email
Password	VARCHAR(100)	Encrypted Password
Role	VARCHAR(20)	Customer/Admin

Description:

The User table stores authentication and role-based access information for all registered users.

9.2 Categories Table

Field Name	Data Type	Description
CategoryID	INT	Primary Key
CategoryName	VARCHAR(100)	Category Name

Description :

The Categories table stores product classifications such as Electronics, Clothing, Books, and Accessories.

9.3 Products Table

Field Name	Data Type	Description
ProductID	INT	Primary Key
ProductName	VARCHAR(100)	Product Name
Price	DECIMAL	Product Price
StockQuantity	INT	Available Quantity
CategoryID	INT	Foreign Key

Description :

The Products table contains detailed information about products available in the system.

9.4 Orders Table

Field Name	Data Type	Description
OrderID	INT	Primary Key
UserID	INT	Foreign Key
OrderDate	DATE	Order Date
OrderStatus	VARCHAR(30)	Current Status

Description :

The Orders table stores information about customer purchases and delivery status.

9.5 Payments Table

Field Name	Data Type	Description
PaymentID	INT	Primary Key
OrderID	INT	Foreign Key
PaymentMethod	VARCHAR(50)	Payment Method
PaymentStatus	VARCHAR(30)	Transaction Status

Description :

The Payments table stores payment transaction details associated with customer orders.

10. Module Description

10.1 User Management Module

This module handles user registration, login, authentication, profile management, and password recovery functionality. It ensures secure access to the system and maintains user information.

10.2 Product Management Module

This module allows administrators to add, update, delete, and organize products. It maintains product information, pricing, categories, and stock availability.

10.3 Shopping Cart Module

This module enables users to add products to their cart, remove products, update quantities, and review selected items before proceeding to checkout.

10.4 Order Management Module

This module processes customer orders, updates order status, generates order confirmations, and provides order tracking functionality.

10.5 Payment Processing Module

This module handles online transactions using various payment methods such as UPI, debit cards, credit cards, and net banking. It communicates with the payment gateway and stores transaction results.

10.6 Inventory Management Module

This module automatically updates stock quantities whenever a purchase occurs and helps administrators monitor inventory levels.

10.7 Report Generation Module

This module generates reports related to sales, inventory, customers, and orders to support business analysis and decision-making.

11. Hardware Requirements

The minimum hardware requirements for running the E-Commerce Management Software are:

- Processor: Intel Core i3 or higher
- RAM: 4 GB minimum
- Storage: 50 GB free disk space
- Network: Stable Internet Connection
- Input Devices: Keyboard and Mouse
- Output Device: Monitor

12. Software Requirements

The software requirements for the system are:

- Operating System: Windows 10/11 or Linux
- Database: MySQL/PostgreSQL
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technologies: Node.js / Python / Java
- Browser: Google Chrome, Mozilla Firefox, Microsoft Edge
- Web Server: Apache/Nginx

13. Security Requirements

1. User passwords shall be stored in encrypted form.
2. All communication shall occur using HTTPS protocols.
3. Access to administrator functions shall be restricted to authorized users only.
4. Payment information shall be securely processed through trusted payment gateways.
5. Regular database backups shall be maintained.
6. The system shall prevent unauthorized access and data modification.

14. Testing

14.1 Test Case 1: User Registration

Input: Valid user information

Expected Result: Successful account creation

Actual Result: Account created successfully

14.2 Test Case 2: User Login

Input: Valid login credentials

Expected Result: User dashboard displayed

Actual Result: Login successful

14.3 Test Case 3: Product Search

Input: Product keyword

Expected Result: List of matching products

Actual Result: Products displayed successfully

14.4 Test Case 4: Order Placement

Input: Valid cart and payment details

Expected Result: Order confirmation generated

Actual Result: Order placed successfully

14.5 Test Case 5: Payment Processing

Input: Valid payment information

Expected Result: Payment completed successfully

Actual Result: Payment approved and recorded

15. Conclusion

ShopSphere provides a complete solution for online shopping by integrating product management, customer management, inventory control, order processing, and payment handling functionalities. The system offers a secure, scalable, and user-friendly environment for customers and administrators. Its modular architecture ensures maintainability and supports future expansion of features and services.

16. Future Scope

Future enhancements planned for ShopSphere include:

1. Development of mobile applications for Android and iOS platforms.
2. Integration of Artificial Intelligence for personalized product recommendations.
3. Chatbot-based customer support services.

4. Multi-language support for international customers.
5. Voice-assisted product search capabilities.
6. Advanced analytics and business intelligence reporting.
7. Integration with multiple international payment gateways.
8. Real-time inventory synchronization and warehouse management.

17. References

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